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U25-E26 (MAT2103SE)

2026

MATHEMATICS

(SEC)

Paper : MAT2103SE

(**LaTeX and HTML**)

Full Marks : 35/20

Time : $1\frac{1}{2}$ / 1 hours

**The figures in the margin indicate
full marks for the questions.**

Special Instructions for Candidates :

1. **Candidates of the Regular Batch (2025) shall answer questions from Section A only.**
2. **Candidates of the Arrear Batch (2024) shall answer questions from Section A and Section B only.**

3. **Candidates of the Arrear Batch (2023)** shall answer questions from **Section A** and **Section B** only.

Important : Candidates must answer questions only from the sections applicable to their respective batch. Answers written from sections not prescribed for their batch shall not be considered for evaluation.

SECTION-A

1. Answer to the following: 1×3=3

(a) What do you mean by preamble in LaTeX document?

LaTeX ডকুমেন্টত preamble বুলিলে কি বুজা যায়?

(b) What is the purpose of `\frac{}{}`?

`\frac{}{}` ৰ উদ্দেশ্য কি?

(c) Which attribute is used in `<img>` to specify image path?

`<img>` টেগত ছবিৰ পথ (image path) দেখুৱাবলৈ কোনটো attribute ব্যৱহাৰ কৰা হয়?

2. Write LaTeX code for :

2

$$\cos \alpha + \cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

তলৰ সমীকৰণটো LaTeX কোডত লিখা

$$\cos \alpha + \cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

3. Answer to the following: **(any three)**

5×3=15

(a) What is the output of `\sqrt{16}`? Write LaTeX code for the following:

(i) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(ii) $x^2 + y^2 = z^2$

`\sqrt{16}` ৰ আউটপুট কি? তলৰ সমীকৰণসমূহৰ বাবে LaTeX কোড লিখা :

(i) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(ii) $x^2 + y^2 = z^2$

- (b) Explain the process of including images in a LaTeX document. Mention required packages and demonstrate with code.

LaTeX ডকুমেন্টত ছবি কেনেকৈ সংযুক্ত কৰা হয় বিৱৰণ দিয়া। প্রয়োজনীয় package উল্লেখ কৰি কোড উদাহৰণ দিয়া।

- (c) What are HTML tags? Give examples of *any four* common tags.

HTML টেগ কি? যিকোনো চাৰিটা সাধাৰণ টেগৰ উদাহৰণ দিয়া।

- (d) Write an HTML program to create a webpage with heading, paragraph, and image.

এটা HTML প্রথাম লিখা য'ত heading, paragraph আৰু image থাকিব।

- (e) What are the different types of lists in HTML? Describe how to include lists in HTML with an example.

HTML ত কিমান প্ৰকাৰৰ তালিকা (lists) আছে? উদাহৰণসহ ব্যাখ্যা কৰা।

SECTION-B

1. (a) What is the use of `<hr>` tag in HTML?

`<hr>` টেগৰ ব্যৱহাৰ কি?

1

2. Answer to the following:

2×2=4

- (a) Name *any one* package in LaTeX and state its use.

LaTeX-ত যিকোনো এটা package ৰ নাম আৰু তাৰ ব্যৱহাৰ লিখা।

- (b) What is the difference between `<h1>` and `<h6>`?

`<h1>` আৰু `<h6>` ৰ মাজত পাৰ্থক্য কি?

3. Answer to the following: (any two)

5×2=10

- (a) Explain the structure of a LaTeX document. Write a sample code illustrating document class, preamble, and body.

LaTeX ডকুমেন্টৰ গঠন ব্যাখ্যা কৰা। document class, preamble আৰু body দেখুৱাই এটা উদাহৰণ কোড লিখা।

- (b) Explain how to create mathematical equations in LaTeX. Write LaTeX code for the following:

$$(x+a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$

LaTeX-ত গণিতৰ সমীকৰণ কেনেকৈ লিখা হয় ব্যাখ্যা কৰা। তলৰ সমীকৰণৰ বাবে LaTeX কোড লিখা :

$$(x+a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$

- (c) Write HTML code to create a table with 4 rows and 3 columns. The column headings should be Roll Number, Name, and Department.

HTML কোড লিখা য'ত 4 টা শাৰী (rows) আৰু 3 টা স্তম্ভ (columns) থকা এখন টেবুল থাকিব।

স্তম্ভসমূহ হ'ব : Roll Number, Name আৰু Department.
